

L8: Accuracy Comparison of Community Detection Algorithms

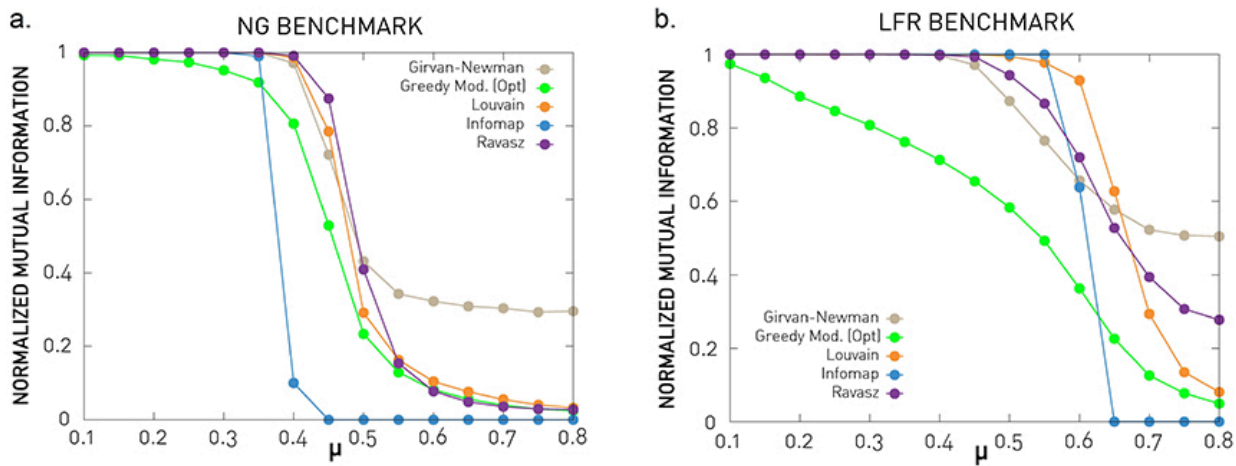


Figure 9.27 from Network Science by Albert-László Barabási

Let's now compare some of the community detection algorithms we have seen so far (at least those that result in non-overlapping communities).

In the previous lesson, we covered the Girvan-Newman algorithm (*top-down hierarchical clustering based on the edge betweenness centrality metric*), the Ravasz algorithm (*agglomerative hierarchical clustering*), the greedy modularity optimization algorithm, and the Louvain algorithm (*also based on modularity optimization*). The plots also show results for Infomap, which is an algorithm that is based on the maximization of an information-theoretic metric of community structure rather than modularity.

The comparisons use the NMI metric, and the two benchmarks we introduced earlier: GN and LFR. Recall that the parameter μ quantifies the strength of the community structure in the network – when $\mu < 0.5$ the network has clearly separated communities, while μ is close to 1 there is no real community structure.

The benchmark parameters are $N=1,000$ nodes, average degree: $\bar{k}=20$, degree exponent (LFR) $\gamma=2$, maximum degree (LFR): $k_{max}=50$, community size exponent (LFR) $\zeta=1$, maximum community size (LFR): 100, minimum community size (LFR): 20. Each curve is averaged over 25 independent realizations.

The Louvain algorithm and the Ravasz algorithm (*cutting the dendrogram based on maximum modularity*) seem to behave consistently well in both benchmarks. The Girvan-Newman algorithms seem to report communities even where there are no real communities (for high values of μ).

Food for Thought

The greedy modularity maximization algorithm is less accurate even for low values of μ , when the communities are quite clearly defined. What do you think causes this inaccuracy?